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|----------------------------------------|---------------------------------------------------|---------------------|-----------------------|
| Type Assessing/ Type of Assessment:    | <b>GA Assignment</b>                              |                     |                       |
| Kwalifikasie/ Qualification:           | <b>B Eng (Industrial)</b>                         |                     |                       |
| Modulekode/ Module code:               | <b>INGB417</b>                                    | Tydsduur/ Duration: | <b>6-13 June 2025</b> |
| Module beskrywing/ Module description: | <b>Facilities Design</b>                          | Maks/ Max:          | <b>100</b>            |
| Eksaminator(e)/ Examiner(s):           | <b>Willemien Kruger (BEng Industrial, M Eng)</b>  |                     |                       |
| Internal Moderator:                    | <b>Marius Alberts (BEng Industrial)</b>           |                     |                       |
| External Examiner:                     | <b>Prof Chris van Schoor (PhD Industrial Eng)</b> |                     |                       |

### **Instructions:**

Read the DESIGN BRIEF as given below, then do appropriate and relevant research to develop your FACILITY DESIGN. This is to be presented in a Facility Design Report (like the Group Assignment report), including all relevant chapters to demonstrate your understanding of the Engineering Design process. An in-person Debrief session will be held on Friday 6 June 13h00-14h00 (in IE boardroom) to clarify expectations and answer any questions you may have.

Your FACILITY DESIGN **Report** must be uploaded (pdf), to eFundi on **Friday 13 June 2025 by 8h00.**

You do not have to submit a printed report.

*The body of the report should be no longer than 40 pages. Appendices may be included and will not count towards the 40 pages. Appendices will only be considered if they are appropriately referred to in the body of the report. No additional files, drawings or spreadsheets will be considered by the examiners.*

*Remember that more words do not necessarily reflect better insight and since this subject is about a Layout plan and the supporting key assumptions and calculations, focus on getting that right.*

No changes to the report will be accepted once it is uploaded to eFundi. No late or resubmissions will be accepted.

Exit interviews will be done with an examiner panel on **Wednesday 18 June (9h00-16h00) as well as Thursday 19 June 2025 (9h00-13h00).** The detail schedule and venue have been communicated in the week of 20 May 2025 for individual planning purposes. Please bring only a colour **PRINTED A2 (or larger if you wish) of the MASTER FACILITY PLAN** to the interview to be part of the discussion.

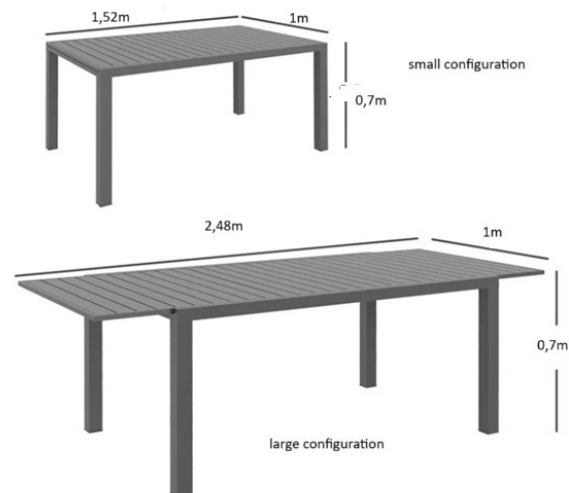
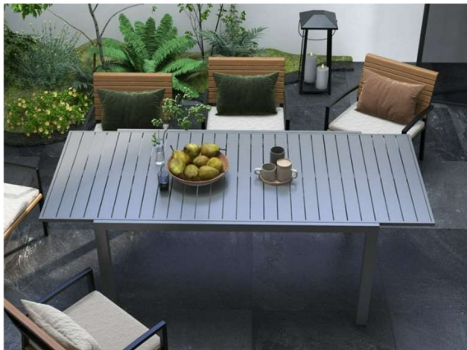
The main purpose of this assignment is to assess your ability to make use of a systematic engineering design approach as the Graduate Attribute (GA) 3 to graduate as an Industrial Engineer. You must obtain at least 50% for the exam report as well as passing GA3. Failure to demonstrate competence in this GA will result in failure of this assessment and the module.

Students are expected to complete the assignment independently, and a declaration will be required when uploading the final report. Any ideas, conversations or content used in the design process must be credited and referenced. The final design must be an original piece of work that the student is solely responsible for developing and presenting.

**DESIGN BRIEF: TIPTOP TABLE MANUFACTURING**

Background – TIPTOP TABLES is a manufacturer of outdoor tables and chairs supplying the local South African market for the past 25 years. Current sales for specifically the outdoor **set** including **one** lightweight 2in1 aluminium outdoor **table** with **four chairs** (to make up a complete set of one table and 4 chairs) have picked up such that a need for a new facility for this product series alone has been identified. This table also has a unique feature in that it can be a small or large table depending on the configuration. Sales forecasts for up to 5 000 per month of these outdoor sets is expected, which cannot be met within their current facilities as this is not the only product made. Management has asked for the plans for a new facility which will help them make strategic decisions on where to build it. You can assume that access to the facility from public roads will be a requirement. The facility must leave about 20% space available for future expansion needs.

TIPTOP requires you, as Industrial Engineer, to apply a systematic engineering design approach to develop a comprehensive FACILITIES LAYOUT for the new **manufacturing** facility. Apart from providing space for the main production processes of manufacturing the tables, the facility needs to accept delivery vehicles/trucks from *suppliers* bringing raw materials to offload, it needs to store all materials inside the facility, as well as provide a place for *customers* collecting orders, or their own trucks loading orders for delivery to customers.

**THE PRODUCT – OUTDOOR SET OF ALUMINIUM 3in1 TABLE and 4 CHAIRS**

These pictures indicate the look and dimensions of the table to be made.

This patio dining table is made of naturally water-safe aluminium alloy with an extra protective powder coat for the charcoal grey colour. Aluminium makes it light to move when needed and even faster at expanding. A slatted surface makes the extendable outdoor dining table easy to wipe dry after rain. It is packaged as partly assembled, as the 4 legs can be assembled at home.

Part of the set include 4 chairs also made from aluminium alloy with the seat and backing made from weaved rattan. The dimensions are not given, as you only need to know that a standard aluminium pipe tube of 6m length is used for the frame. The rattan seat/backing is outsourced to another company which take away the frames for their work daily and bring back completed chairs daily.



## PRODUCT AND MANUFACTURING INFORMATION

The easiest way to understand the manufacturing process is to watch the current process where a variety of outdoor furniture is made: <https://www.youtube.com/watch?v=b3m4THLU2lw>

The manufacturing of the chairs can be viewed here:

<https://www.youtube.com/shorts/Je6bNHXVILQ>

*[please note that the video is for background and information purposes only]*

In the new facility they will be doing things similarly, but they may need to purchase more machines and get more labour to fulfil demand. You are required to help management with these calculations on work and capacity demand. The only raw materials bought are aluminium tubes from <https://www.aluminiumtrading.co.za/>

*Rectangular tubes* (40mmx20mmx1,2mm) are bought to make the frame for the tabletop as well as the inserted structure that allows the two tabletops to slide together for the small configuration.

*Flat bars* (101,6mmx9,52mm) are bought for the tabletop (this is welded to a frame).

*Square tubes* (60mmx60mm) are bought to make the table legs.

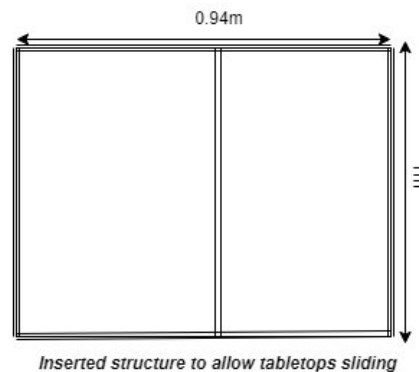
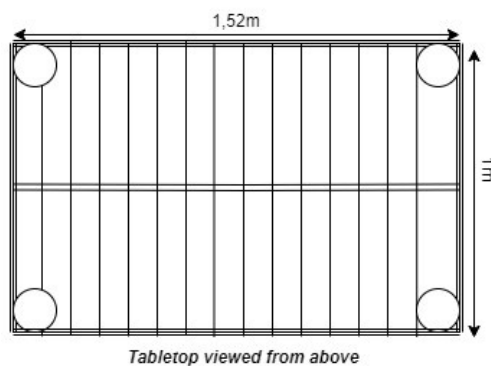
*Round tubes* (25,4mmx1,6mm) are bought for making the chairs.

The main processes to focus on are specified as follows:

| <i>For the TABLE manufacturing:</i>                                                                                  | <i>Estimated processing times</i> |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| CUTTING of the standard 6m rectangular tube for making the table frame (this is done by a CNC laser cutting machine) | 1min for all parts cut per table  |
| WELDING (automated robot) to make the structure for enabling 2 tabletops to slide into each other                    | 5min per sliding structure        |
| WELDING (manual) of the tabletop frame to the flat bars to make the tabletops (2 tops are needed to make 1 table)    | 10min per one tabletop frame      |
| GRINDING and POLISHING is done by hand to just ensure smooth finishes                                                | 1min per tabletop                 |

|                                                                                                                                                                |                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| POWDER COATING only with machine in a chamber specially design for this activity                                                                               | 30sec per structure                                                                                                              |
| ASSEMBLY by hand to put two tabletops together with the sliding structure to make the 3in1 table. 20 screws are used per table for this activity.              | 5min per table                                                                                                                   |
| Just before packing the legs, rubber levelling feet are put on which are bought from outside.                                                                  | 1min per 4 legs                                                                                                                  |
| PACKAGING by hand to ensure quality check and correct orders are packed. Packaging is done to a maximum of 5 tables together with the legs, in one carton box. | 5min per set.<br>Small orders can be collected – less than 5 sets<br>Large orders are delivered to customers – more than 5 sets. |

The specifications of the table frame are as follows:



|                                                                                                                                                                                                                                               |                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <p>Tabletop:</p> <p>Viewed from beneath, fifteen flat bars are laid out next to each other and then welded to three pieces of 1,52m in length as well as two pieces of 1m in length. Legs are put into the circular holds at the corners.</p> | <p>Inserted structure for sliding:</p> <p>Three pieces of 1m in length are welded together with two pieces of 0,94m.</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|

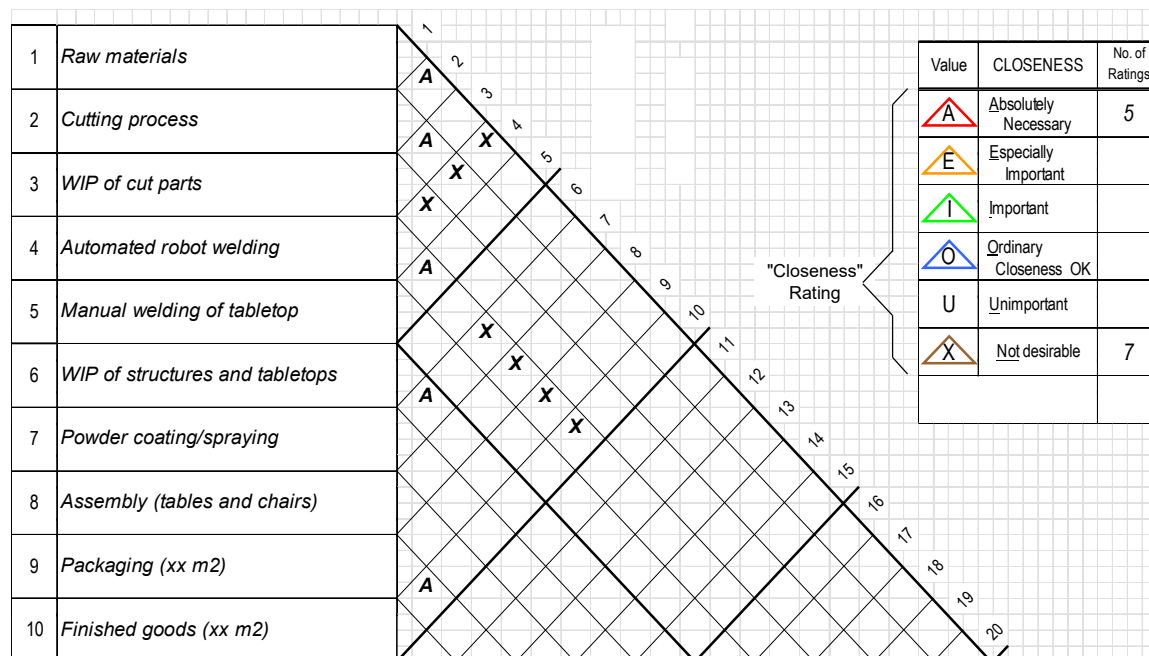
| <i>For the CHAIRS manufacturing:</i>                                                                                                | <i>Processing time estimates</i> |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| CUTTING of the round tube for the frame. A standard 6m tube will be cut into 5 pieces for the chair frame by a CNC cutting machine. | 1min per tube                    |
| BENDING of the tubes to make the frame parts                                                                                        | 3min per chair                   |
| WELDING (manual) of the tubing to make the frame                                                                                    | 5 min per chair                  |

|                                                                                                                                                                                                                                               |                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Putting on the Rattan is outsourced.                                                                                                                                                                                                          | The supplier delivers 1000 chairs per day, which is sufficient for meeting demand. |
| STACKING of chairs to await combining it with the tabletop to make a complete set as a finished product. Chairs are not packaged but just stacked together with tiestraps as associated with the packed tables in sets of 5 for large orders. | 10min to stack 20 chairs                                                           |

You should design enough space for any work-in-progress needed between processes. Also remember to make provision for scrap that may be made at the CUTTING stations.

Some specific requirements management needs are to have a separate space assigned to any welding work done, and allow safety space where the industrial robots are working. If any other details are needed to determine space requirements, you can make assumptions. Process flow charts must form part of their Quality Management System. You may use other tools and techniques for better application to design a Facility Layout.

Some limitations have been set by shopfloor workers upon workstations arrangement as follows due to movement safety as well as ventilation issues. These specifications must be considered when determining the layout. It may also be useful to complete the chart to help in planning the layout. The below activity relationship chart is applicable to tables.



Other policies to consider include:

### **Manufacturing Operating Times**

The company currently operates a 9 hour shift per day, 5 days per week. Operators are given 30 minutes for lunch and two 15 minute tea breaks per day. You can assume an average of 20 working days per month. Overtime or a Saturday shift is possible but should only be considered to provide flexibility in capacity.

### **Inventory Levels**

Inventory level policies are as follows:

Raw materials and components: 2 weeks stock     Finished goods: 1 month/4 weeks stock as they are shipping overseas and need to ship big batches at a time.

Your own assumptions can be made regarding Work-In-Process (WIP) inventory at different stages of production and the space needed on the shop floor.

### **Deliveries and Collections**

Raw material and component deliveries take place throughout the week. Finished good collections take place daily, or when taking bulk orders TIPTOP use their own fleet. All deliveries and collections will take place during production hours.

## **FACILITY DESIGN EXPECTATIONS**

The following provides detail expected to see in the Final Design.

### **• Materials Receiving and Storage Design considerations**

- Clear *receiving space* for all suppliers,
- *storage* of all raw materials and components including details of 3D space requirements based on calculations, material flow and layout and material handling equipment recommendations.
- *Staging* of materials and components at manufacturing and assembly.

### **• Manufacturing Operations Design**

- *Process flow charts* to indicate clear understanding of the manufacturing processes required, including insight into bottlenecks, workstations and departments.
- *Production capacity analysis* (including detail calculations) compared to forecast demand to improve on process design so that the manufacturing processes can meet demand. Some changes will have to be made to meet future demand.
- *Capacity analysis for all manufacturing processes* including machine and personnel requirements organized in workstations.
- *Recommended machinery* for all critical manufacturing processes.
- Operator and material movement requirements including movement between manufacturing processes and movement of material from raw material storage to the WIP in manufacturing areas and then to finished goods storage. This can be summarised as flow and should be indicated on the final layout.
- Detailed layout and space planning for all workstations and WIP stations where required.
- Interface design for dispatch and receiving of chairs for external rattan seat/backing.

### **• Finished Product Storage and Shipping Operations Design**

- Finished product and order storage capacity including details of 3D space requirements based on calculations, flow and layout and equipment and materials handling recommendations.

- Clear and logical dispatch operation space for customers getting their small orders or
- Shipping operations where the company has their own trucks taking large orders to distribution centres.
- **Supporting Facilities & Services Design**
  - Personnel facilities for all employees on site.
  - Offices and related facilities for: Management (1 senior and 2 middle management), administrative (5 people including Purchasing and Finance and Sales) and operational staff (as you have determined including supervisors).
  - Overall security considerations, entrance, exit and passenger and goods vehicle flow on-site including parking.
  - Waste handling and management.
  - Utility requirements including water, electricity, plumbing etc.

## THE DELIVERABLES

This project requires you to apply a systematic engineering design approach to develop a **comprehensive** facilities layout (Master Facility Plan) that identifies and meets the needs of the client.

TIPTOP has specifically asked you to include the following in your report:

1. A set of detailed facility design requirements.
2. A thorough capacity/demand analysis to clearly demonstrate production capacity needs.
3. At least three alternative design solutions. All considerations must be communicated.
4. A detailed justification of chosen design(s) including evaluation criteria.
5. A detailed master facilities plan that includes sufficient details of workstation, material handling, storage, receiving, dispatch and personnel space requirements to present to management.
6. A well-prepared Facility Design Report.

The report must include comprehensive evidence to indicate that you followed a systematic Engineering Design process and summarise your key assumptions and capacity calculations.

An opportunity to explain how you approached this project will be given in an EXIT INTERVIEW where you will have 10 minutes to explain how you approached the design problem, what you have done practically to develop design options, and explain the logic of your Master Facility Plan. This is not a presentation, but more question and answers session with examiners to test insight. Please bring with the 2D layout colour printed in A2 format to use as reference.

Attached the Rubric that will be used for marking each student individually.

| Facilities Design – INGB417 - Examiner mark sheet for use by internal and external examiner 2025 |                    |                |
|--------------------------------------------------------------------------------------------------|--------------------|----------------|
| Student Particulars                                                                              |                    |                |
| Surname                                                                                          | Initials           | Student number |
|                                                                                                  |                    |                |
| Examiner Particulars                                                                             |                    |                |
| Internal or External (circle)                                                                    | Initials & Surname | Affiliation    |
|                                                                                                  |                    |                |

| GA 3: Engineering Design                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Design</b> creative solutions for complex engineering problems and design systems, components or processes to meet identified needs. |

| EVALUATION CRITERIA                                                                                                                                                                           |           |                  |               |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------|---------------|--------------|
|                                                                                                                                                                                               | Weighting | Mark (Interview) | Mark (Report) | Mark (Final) |
| Clear interpretation of the design problem and <b>formulation/calculation of design requirements or specifications.</b> (includes appropriate use of <b>analytical tools</b> and techniques.) | /30       |                  |               |              |
| Practical <b>insight shown</b> during the design process and in the design solution                                                                                                           | /20       |                  |               |              |
| Sufficient <b>detail provided in the solution</b> to meet the design requirements (Receiving & Dispatch, Processes, Storage, Offices, Support functions all in MFP)                           | /30       |                  |               |              |
| Effective use of a structured and rigorous <b>design process</b> (Report structure corresponding to process steps)                                                                            | /10       |                  |               |              |
| Effective <b>communication</b> of the design process and solution                                                                                                                             | /10       |                  |               |              |
| <b>FINAL MARK (Interview 50: Report 50)</b>                                                                                                                                                   | /100      | Interview mark   | Report total  | Total mark   |

| ASSESSMENT OF GA 3 (ENGINEERING DESIGN) |            |
|-----------------------------------------|------------|
| GA Achieved?<br>YES/NO                  |            |
| Comments:                               |            |
|                                         |            |
| Assessment Date:                        | Signature: |